

Outline of Comments: FDA Draft Guidance - *Human Cells, Tissues, and Cellular and Tissue-Based Products (HCT/Ps) from Adipose Tissue: Regulatory Considerations*

FDA Guidance Provision	Comment	Rationale
Draft guidance reportedly issued because “FDA has recently received numerous inquiries regarding HCT/Ps manufactured from adipose tissues.”	Most of those inquiries likely involved discovery and use of mesenchymal stromal cells (MSCs) which are located within the structure of adult fat tissue. The primary purpose of the guidance seems directed specifically to “slam the door” concerning the use of MSCs without development of a full drug/biologic application under §351 PHSA. Issuance of such guidance pushes U.S. patients further behind in realizing the therapeutic value of MSCs in safely treating disease. Such unreasonably narrow definitions also ignore existing science and strip the Agency of an important regulatory tool under §361 PHSA used so effectively to use stem cells to treat blood-borne disease. This FDA position will establish a precedent that will affect MSCs derived from all other tissue, such as bone marrow, cord tissue, amniotic fluid, etc. It would be more scientifically valid to consider the methods of manufacturing	BPC and many clinical experts involved in regenerative medicine believe that the mechanisms used by FDA for new drug development are: (1) unnecessary to guarantee safe and effective use, (2) outdated; and (3) not tailored to the use of adult stem cells.

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	MSCs derived from these tissues and assigning the regulatory pathway accordingly. Autologous MSC cells derived from a direct processing method using FDA approved reagents, and administered in the same surgical procedure, or within a few days of initiation of manufacturing, should be regulated under 361. Autologous or allogeneic cells manufactured with complex methods (e.g., more than minimal manipulation) should be regulated under 351.	
Processing to isolate non-adipocyte or non-structural components from adipose tissue (with or without subsequent cell culture or expansion) is generally considered more than minimal manipulation—because the adipose tissue are entirely removed from the non-adipocyte or non-structural isolates, thereby altering the original relevant characteristics relating to the tissue’s utility for reconstruction, repair, or replacement (Guidance at pg. 3).	In adopting this narrow interpretation, FDA is deviating unnecessarily from the plain meaning of its own 1271 regulation. In the definition of “minimal manipulation” (§1271.3(f)), FDA properly focused on whether the intended product was a cell or the underlying structural tissue. Only if the focus was on the adipose tissue itself, for example, then the requirements of §1271.3(f)(1) applied “that the processing should not alter the original relevant characteristics of the tissue relating to the tissue’s utility for reconstruction, repair, or replacement.” However, where the intended product is	When processed under the other limitations of the HCT/P regulatory framework in 1271, the use of these stem cells is safe and effective (e.g., minimally manipulated, homologous use, no combination with other prohibited articles, autologous use, or allogeneic in relative, etc.). FDA should follow its own regulations without linking MSCs to the structural adipose tissue in which they are located and isolated.

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	the MSC/ stromal stem cells, §1271.3(f)(2) applies related to cells, not §1271.3(f)(1) related only to structural tissue. The cell provision in §1271.3(f)(2) applies a different standard; that the processing “does not alter the relevant biological characteristics of the cells.” If those MSCs have the biological ability to differentiate and proliferate, it should be minimal manipulation to isolate them and allow them to use these inherent biological characteristics.	
It is more than minimal manipulation to isolate adipose-derived stromal/stem cells from adipose tissue for clinically therapeutic uses. This processing breaks down and eliminates the structural components of this tissue that function to provide cushioning and support thereby altering the original relevant characteristics of the HCT/P relating to its utility for reconstruction, repair and replacement. (Guidance at pg. 4)	As noted above, this cited criteria is limited to use of the structural adipose tissue itself, and not MSC cells isolated from this tissue, that are the intended product for human use.	Rather than using non-binding guidance to prevent the use of stem cells in regenerative medicine under 1271, FDA should acknowledge the state of the science in this regard and permit this narrow use in these safest of circumstances.
In contrast, processing that does not affect the adipose tissue’s utility as a structural tissue for reconstruction, replacement, or repair may be	This example should be limited to §1271.3(f)(1) when the structural adipose tissue itself is the product intended for patient use, or when a cell-containing	

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considered minimal manipulation (e.g., aliquoting, rinsing, removal of macroscopic debris, and freezing). (Guidance at pg. 4)	tissue is the intended product.	
Use of adipose tissue to cosmetically fill void in the subcutaneous space in the face or hands is homologous use because of its use to support the subcutaneum.	This obvious and narrow example of homologous use should not preclude use of MSCs isolated from adipose tissue if the tissue itself is not also retained for therapeutic use. The homologous definition focuses on whether the cells, or tissue, “perform the same basic function in the recipient as in the donor.”	
Breast augmentation would generally be considered non-homologous use because the basis function of breast tissue is to produce milk (lactation) after childbirth.	Use of adipose tissue and MSCs for breast reconstruction is now a common clinical practice. When considering that the breast is largely composed of fat tissue, and that the breast has many functions other than lactation, applying fat-based HCT/Ps to restore breast shape should be considered homologous use.	
If stem cells from the lipoaspirate are isolated, the adipose tissue would no longer be considered HCT/P. Thus, even if these cells are injected into the same patient from which they are removed during the same surgical procedure, the	FDA confuses use of the isolated cells themselves from use of the underlying tissue in which these cells are harvested. MSCs are unique in that they do not undergo lineage differentiation in fat tissue. They may have another function in	The plain language of FDA’s definitions of minimally manipulated and homologous contained in §1271.3(c) and (f) allow use of cells for purposes apart from the function of the tissue in which they are located.

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exception under §1271.15(b) [for use during the same surgical procedure] would not apply. (Guidance at pg. 8)	this tissue, but it is currently unknown. Yet, they possess the biologic characteristics to reproduce and differentiate into other types of cells if isolated and relocated throughout the body. They also act as effector/signaling cells exerting anti-inflammatory and immune tolerance effects in vivo. These uses should be considered a homologous use of these cells.	